UNIT-IV

FOURIER TRANSFORMS

Topic Learning Objectives:
Upon Completion of this unit, students will be able to:

- Understand the Fourier integral theorem.
- Apply complex Fourier transform, Fourier sine and cosine transforms.
- Demonstrate - the properties of transforms.
- Understand and apply Convolution theorem and Parseval's identity.

Introduction:
The theory of Fourier transforms afford mathematical devices through which solutions of numerous boundary value problems of engineering can be obtained. For example conduction of heat, transverse vibrations of a string, transverse oscillations of an elastic beam, free and forced vibrations of a membrane, transverse oscillations of an elastic beam, free and forced vibrations of a membrane, transmission lines.

Fourier Transform is a technique employed to solve ODE's, PDE's and Integral equations. The choice of particular transform to be employed for the solution of an equation depends on the boundary conditions of the problem. A Fourier transform when applied to partial differential equation reduces the number of independent variables involved by one.

FOURIER INTEGRAL THEOREM:

Consider a function $f(x)$ which satisfies the Dirichlet's conditions in every interval $(-c,c)$, then

$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi x}{c}+b_n\sin\frac{n\pi x}{c}\right) \quad \ldots\ldots \tag{1}$$

$$a_0=\frac{1}{c}\int_{-c}^{c}f(t)dt\,,$$

$$a_n=\frac{1}{c}\int_{-c}^{c}f(t)\cos\left(\frac{n\pi}{c}\right)tdt, \qquad n=1,2,3,\ldots..$$

$$b_n=\frac{1}{c}\int_{-c}^{c}f(t)\sin\left(\frac{n\pi}{c}\right)tdt, \qquad n=1,2,3,\ldots\ldots$$

substituting the values of a_0, a_n, b_n in (1),

$$f(x)=\frac{1}{2c}\int_{-c}^{c}f(t)dt,+\frac{1}{c}\sum_{n=1}^{\infty}\int_{-c}^{c}f(t)\cos\frac{n\pi(t-x)}{c}dt\,, \tag{2}$$

Assuming $\int_{-\infty}^{\infty}|f(x)|dx$ converges, as $c\to\infty$ equation (2) becomes

$$f(x)=\frac{1}{\pi}\int_{0}^{\infty}\int_{-\infty}^{\infty}f(t)\cos\lambda(t-x)dtd\lambda,$$

which is known as Fourier integral of $f(x)$.

FOURIER SINE AND COSINE INTEGRALS:

Expanding $\cos\lambda(t-x)$, if $f(x)$ is an odd function, $f(t)\cos\lambda t$ is odd function, $f(t)\sin\lambda t$ is even, equation (2) reduces to

$$\boldsymbol{f(x) = \frac{2}{\pi}\int_0^\infty \sin\lambda x \int_0^\infty f(t)\sin\lambda t dt d\lambda}$$

which is known as Fourier sine integral.

Similarly, if $f(x)$ is even function, equation (2) reduces to

$$\boldsymbol{f(x) = \frac{2}{\pi}\int_0^\infty \cos\lambda x \int_0^\infty f(t)\cos\lambda t dt d\lambda,}$$

which is known as Fourier cosine integral.

Example:

1. Express $\boldsymbol{f(x) = \begin{cases} 1 \ for \ 0 \le x \le \pi \\ 0 \ for \ x > \pi, \end{cases}}$ as a Fourier sine integral and hence evaluate

$$\boldsymbol{\int_0^\infty \frac{1-\cos(\pi\lambda)}{\lambda}\sin(x\lambda)d\lambda.}$$

Solution: The Fourier Sine integral for

$$f(x) = \frac{2}{\pi}\int_0^\infty \sin(\lambda x)d\lambda \int_0^\infty f(t)\sin(\lambda t)dt,$$

$$= \frac{2}{\pi}\int_0^\infty \sin\lambda x d\lambda \int_0^\pi \sin(\lambda t)dt,$$

$$= \frac{2}{\pi}\int_0^\infty \frac{1-\cos(\pi\lambda)}{\lambda}\sin(x\lambda)d\lambda,$$

$$\int_0^\infty \frac{1-\cos(\pi\lambda)}{\lambda}\sin(x\lambda)d\lambda = \frac{\pi}{2} f(x) = \begin{cases} \frac{\pi}{2} \ for \ 0 \le x < \pi \\ 0 \ for \ x > \pi \end{cases}.$$

At $x = \pi$, which is a point of discontinuity of $f(x)$, The value of the above integral is $\frac{\pi}{4}$.

Fourier Transform:

Complex Fourier Transform or Fourier Transform of $f(x)$ is given by

$$\hat{f}(\alpha) = F[f(x)] = \int_{-\infty}^{\infty} f(x) e^{i\alpha x} dx,$$

provided the integral exists.

Also the function $f(x)$ given by

$$\boldsymbol{F^{-1}[\hat{f}(\alpha)] = f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\alpha)\, e^{-i\alpha x} d\alpha,}$$

is called inverse Fourier transform of $\hat{f}(\alpha)$.

Note: The function $f(x)$ is said to be self-reciprocal with respect to Fourier transform if $\hat{f}(\alpha) = f(\alpha)$.

Fourier sine and cosine Transforms:

$$\hat{f}_s(\alpha) = F_s[f(x)] = \int_0^{\infty} f(x) \sin \alpha x \, dx,$$

is called Fourier sine transform of $f(x)$.

$$F_s^{-1}\left[\hat{f}_s(\alpha)\right] = f(x) = \frac{2}{\pi}\int_0^{\infty} \hat{f}_s(\alpha) \sin \alpha x \, d\alpha,$$

is called Inverse Fourier sine transform of $\hat{f}_s(\alpha)$.

Similarly

$$\hat{f}_c(\alpha) = F_c[f(x)] = \int_0^{\infty} f(x) \cos \alpha x \, dx,$$

is called Fourier cosine transform of $f(x)$.

$$F_c^{-1}\left[\hat{f}_c(\alpha)\right] = f(x) = \frac{2}{\pi}\int_0^{\infty} \hat{f}_c(\alpha) \cos \alpha x \, d\alpha,$$

is called Inverse Fourier cosine transform of $\hat{f}_c(\alpha)$.

Relation between Fourier Sine and Cosine Transform:

(i) $F_s[xf(x)] = -\frac{d}{ds} F_c[f(x)]$ and

(ii) $F_c[xf(x)] = \frac{d}{ds} F_s[f(x)]$

Properties of Fourier Transforms:

Following are some basic properties of Fourier Transforms:

1. **Linearity Property:**
For any two functions $f(x)$ and $\phi(x)$ (whose Fourier Transforms exist) and any two constants a and b,

$$F[af(x)+b\phi(x)] = aF[f(x)]+bF[\phi(x)].$$

Proof
By definition,

$$F[af(x)+b\phi(x)] = \int_{-\infty}^{\infty}[af(x)+b\phi(x)]e^{i\alpha x}dx,$$
$$= a\int_{-\infty}^{\infty} f(x)e^{i\alpha x}dx + b\int_{-\infty}^{\infty}\phi(x)e^{i\alpha x}dx,$$
$$= aF[f(x)]+bF[\phi(x)],$$

This is the desired property.
In particular, if $a = b = 1$,

$$F[f(x)+\phi(x)] = F[f(x)]+F[\phi(x)].$$

Again if $a = -b = 1$,

$$F[f(x)-\phi(x)] = F[f(x)]-F[\phi(x)].$$

2. **Change of Scale Property**:
If $\hat{f}(\alpha) = F[f(x)]$, then for any non - zero constant a,

$$F[f(x)] = \frac{1}{|a|}\hat{f}\left(\frac{\alpha}{a}\right).$$

<u>Proof</u>: By definition,

$$F[f(ax)] = \int_{-\infty}^{\infty}[f(ax)]e^{i\alpha x}dx. \qquad (1)$$

<u>Suppose $a > 0$</u>: Set $ax = u$. Then expression (1) becomes

$$F[f(ax)] = \int_{-\infty}^{\infty}[f(u)]e^{i\left(\frac{\alpha}{a}\right)u}\frac{du}{a},$$
$$= \frac{1}{a}\hat{f}\left(\frac{\alpha}{a}\right). \qquad (2)$$

<u>Suppose $a < 0$</u>. Set again $ax = u$, then (1) becomes

$$F[f(ax)] = \int_{\infty}^{-\infty}[f(u)]e^{i\alpha\left(\frac{u}{a}\right)}\frac{du}{a},$$
$$= -\frac{1}{a}\int_{-\infty}^{\infty}[f(u)]e^{i\left(\frac{\alpha}{a}\right)u}du,$$
$$= -\frac{1}{a}\hat{f}\left(\frac{\alpha}{a}\right). \qquad (3)$$

Expressions (2) and (3) may be combined as

$$F[f(ax)] = \frac{1}{|a|}\hat{f}\left(\frac{\alpha}{a}\right).$$

This is the desired property.

3. **Shifting Properties:**

For any real constant 'a',

$(i)\quad F[f(x-a)] = e^{i\alpha a}\hat{f}(\alpha),$

$(ii)\quad F[e^{iax}f(x)] = \hat{f}(\alpha+a).$

Proof:

(i) $F[f(x)] = \hat{f}(\alpha) = \int_{-\infty}^{\infty}[f(x)]e^{i\alpha x}dx,$

Hence, $F[f(x-a)] = \int_{-\infty}^{\infty}[f(x-a)]e^{i\alpha x}dx,$

Set $x\text{-}a = t$. Then $dx = dt$.

Then,

$$F[f(x-a)] = \int_{-\infty}^{\infty}[f(t)]e^{i\alpha(t+a)}dt,$$
$$= e^{i\alpha a}\ \hat{f}(\alpha).$$

ii) $\hat{f}(\alpha+a) = \int_{-\infty}^{\infty} f(x)\,e^{i(\alpha+a)x}dx,$

$$= \int_{-\infty}^{\infty}\left[f(x)e^{iax}\right]e^{i\alpha x}dx,$$
$$= \int_{-\infty}^{\infty} g(x)e^{iax}\ dx,\quad where\ g(x) = f(x)e^{iax},$$
$$= F[g(x)],$$
$$= F[e^{iax}f(x)].$$

This is the desired result.

4. **Modulation Property:**

If $F[f(x)] = \hat{f}(\alpha),$

then, $F[f(x)\cos ax] = \frac{1}{2}\left[\hat{f}(\alpha+a) + \hat{f}(\alpha-a)\right]$ where 'a' is a real constant.

Proof :

$$\cos ax = \frac{e^{iax}+e^{-iax}}{2},$$

$$F[f(x)\cos ax] = F\left[f(x)\left(\frac{e^{iax}+e^{-iax}}{2}\right)\right],$$
$$= \frac{1}{2}\left[\hat{f}(\alpha+a) + \hat{f}(\alpha-a)\right], \text{by using linearity and shift properties.}$$

This is the desired result.

Note: Similarly

$$F[f(x)\sin ax] = \frac{1}{2}\left[\hat{f}(\alpha+a) - \hat{f}(\alpha-a)\right].$$

Examples:

1. Find the Fourier Transform of the function $f(x)=e^{-a|x|}$ where $a>0$.

Solution: For the given function,

$$F[f(x)]=\int_{-\infty}^{\infty}e^{-a|x|}e^{i\alpha x}\,dx,$$

$$=\left[\int_{-\infty}^{0}e^{-a|x|}e^{i\alpha x}\,dx+\int_{0}^{\infty}e^{-a|x|}e^{i\alpha x}\,dx\right],$$

Using the fact that $|x|=x, 0\le x<\infty$ and $|x|=-x, -\infty<x\le 0,$

$$F[f(x)]=\left[\int_{-\infty}^{0}e^{ax}e^{i\alpha x}\,dx+\int_{0}^{\infty}e^{-ax}e^{i\alpha x}\,dx\right],$$

$$=\left[\int_{-\infty}^{0}e^{(a+i\alpha)x}\,dx+\int_{0}^{\infty}e^{-(a-i\alpha)x}\,dx\right],$$

$$=\left[\left\{\frac{e^{(a+i\alpha)x}}{(a+i\alpha)}\right\}_{-\infty}^{0}+\left\{\frac{e^{-(a-i\alpha)x}}{-(a-i\alpha)}\right\}_{0}^{\infty}\right],$$

$$=\left[\frac{1}{(a+i\alpha)}+\frac{1}{(a-i\alpha)}\right],$$

$$=\left[\frac{2a}{(a^2+\alpha^2)}\right].$$

2. Find the Fourier Transform of the function

$$f(x)=\begin{cases}1, & |x|\le a\\ 0, & |x|>a\end{cases},$$

where 'a' is a positive constant. Hence evaluate

(i) $\displaystyle\int_{-\infty}^{\infty}\frac{\sin\alpha a\ \cos\alpha x}{\alpha}\,d\alpha$

(ii) $\displaystyle\int_{0}^{\infty}\frac{\sin\alpha}{\alpha}d\alpha$

Solution: For the given function,

$$F[f(x)]=\left[\int_{-\infty}^{\infty}f(x)e^{i\alpha x}dx\right],$$

$$=\left[\int_{-\infty}^{-a}f(x)e^{i\alpha x}dx+\int_{-a}^{a}f(x)e^{i\alpha x}dx+\int_{a}^{\infty}f(x)e^{i\alpha x}dx\right],$$

$$=\left[\int_{-a}^{a}e^{i\alpha x}dx\right],$$

$$=2\left[\frac{\sin\alpha a}{\alpha}\right].$$

Thus $F[f(x)]=\hat{f}(\alpha)=2\left(\dfrac{\sin\alpha\ a}{\alpha}\right)$. Inverting $\hat{f}(\alpha)$ by employing inversion formula,

$$f(x)=\frac{1}{2\pi}\int_{-\infty}^{\infty}2\left[\ \frac{\sin\alpha a}{\alpha}\ \right]e^{-i\alpha x}d\alpha,$$

$$= \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sin \alpha a(\cos \alpha x - i \sin \alpha x)}{\alpha} d\alpha,$$

$$= \frac{1}{\pi}\left[\int_{-\infty}^{\infty} \frac{\sin \alpha a(\cos \alpha\, x)}{\alpha} d\alpha - i\int_{-\infty}^{\infty} \frac{\sin \alpha\, a \sin \alpha x}{\alpha} d\alpha\right].$$

Here, the integrand in the first integral is even and the integrand in the second integral is odd. Hence using the relevant properties of integral gives

$$f(x) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sin \alpha a \cos \alpha x}{\alpha} d\alpha.$$

or

$$\int_{-\infty}^{\infty} \frac{\sin \alpha a \cos \alpha x}{\alpha} d\alpha = \pi f(x),$$

$$= \begin{cases} \pi, & |x| \le a \\ 0, & |x| > a \end{cases}$$

For $x = 0,\ a = 1$, this yields

$$\int_{-\infty}^{\infty} \frac{\sin \alpha}{\alpha} d\alpha = \pi.$$

Since the integrand is even,

$$2\int_{0}^{\infty} \frac{\sin \alpha}{\alpha} d\alpha = \pi.$$

or

$$\int_{0}^{\infty} \frac{\sin \alpha}{\alpha} d\alpha = \frac{\pi}{2}.$$

3. Find the Fourier Transform of $f(x) = e^{-a^2x^2}$ where 'a' is a positive constant.

Deduce that $f(x) = e^{-x^2/2}$ is self reciprocal with respect to Fourier Transform.

Solution:

$$F[f(x)] = \int_{-\infty}^{\infty} e^{-a^2x^2} e^{i\alpha x} dx,$$

$$= \int_{-\infty}^{\infty} e^{-(a^2x^2 - i\alpha x)} dx,$$

$$= \int_{-\infty}^{\infty} e^{-\left[\left(ax - \frac{i\alpha}{2a}\right)^2 + \frac{\alpha^2}{4a^2}\right]} dx,$$

$$= e^{-\left(\alpha^2/4a^2\right)} \int_{-\infty}^{\infty} e^{-\left(ax - \frac{i\alpha}{2a}\right)^2} dx.$$

Setting $t = ax - \dfrac{i\alpha}{2a}$,

$$F[f(x)] = e^{-\left(\alpha^2/4a^2\right)} \int_{-\infty}^{\infty} e^{-t^2} \frac{dt}{a},$$

$$= \frac{1}{a} e^{-\left(\alpha^2/4a^2\right)} 2\int_{0}^{\infty} e^{-t^2} dt,$$

$$= \frac{1}{a} e^{-\left(\alpha^2/4a^2\right)} \sqrt{\pi}, \text{using gamma function.}$$

$$\hat{f}(\alpha) = \frac{\sqrt{\pi}}{a} e^{-\left(\alpha^2/4a^2\right)}.$$

This is the desired Fourier Transform of $f(x)$.

For $a^2 = 1/2$ in $f(x) = e^{-a^2x^2}$, $f(x) = e^{-x^2/2}$ and $\hat{f}(\alpha) = \sqrt{2\pi}\, e^{-\alpha^2/2}$

Also putting $x = \alpha$ in $f(x) = e^{-x2/2}$, $f(\alpha) = e^{-\alpha 2/2}$. Hence, $f(\alpha)$ and $\hat{f}(\alpha)$ are same but for constant multiplication by $\sqrt{2\pi}$.

Thus $f(\alpha) = \hat{f}(\alpha)$. It follows that $f(x) = \mathrm{e}^{-x^2/2}$ is self reciprocal.

4. Find the Fourier cosine transform of $f(x) = e^{-x^2}$.

Solution: Fourier cosine transform of $f(x) = e^{-x^2}$ is

$$\hat{f}_c(\alpha) = F_c[f(x)] = \int_0^\infty e^{-x^2} \cos \alpha x dx = I.$$

Differentiating under the integral sign w.r.t α

$$\frac{dI}{d\alpha} = \int_0^\infty x e^{-x^2} \sin \alpha x dx = -\frac{\alpha}{2} I,$$

$$\frac{dI}{I} = \int \frac{\alpha}{2} d\alpha + \log c,$$

$$\log I = -\frac{\alpha^2}{4} + \log c = \log(ce^{\frac{-s^2}{4}}),$$

or

$$I = ce^{\frac{-s^2}{4}} \text{ or} \int_0^\infty e^{-x^2} \cos \alpha x dx = ce^{\frac{-s^2}{4}},$$

Putting $\alpha = o, c = \frac{\sqrt{\pi}}{2}$ i.e., $I = \frac{\sqrt{\pi}}{2} e^{\frac{-\alpha^2}{4}}$.

Hence Fourier cosine transform of $f(x) = e^{-x^2}$ is $\frac{\sqrt{\pi}}{2} e^{\frac{-\alpha^2}{4}}$.

5. Find the Fourier sine transform of $f(x) = e^{-|x|}$. Hence show that

$$\int_0^\infty \frac{x \sin mx}{1+x^2} dx = \frac{\pi}{2} e^{-m}, m > 0.$$

Solution: Fourier sine transform of $f(x) = e^{-|x|}$ is

$$\hat{f}_s(\alpha) = F_s[f(x)] = \int_0^\infty e^{-|x|} \sin \alpha x dx = \frac{\alpha}{1+\alpha^2}.$$

Using inverse Fourier Transform

$$f(x)=e^{-x}=\frac{2}{\pi}\int_0^\infty \frac{\alpha}{1+\alpha^2}\sin\alpha x\,d\alpha=\frac{\alpha}{1+\alpha^2}.$$

Changing x to m,

$$\int_0^\infty \frac{x\sin mx}{1+x^2}dx=\frac{\pi}{2}e^{-m}, m>0.$$

Note:

$$F_s[x\,f(x)]=-\frac{d}{d\alpha}\{F_c(\alpha)\};F_c[x\,f(x)]=\frac{d}{d\alpha}\{F_s(\alpha)\}.$$

6. Find the Fourier sine transform of xe^{-ax}.

Solution:

$\boldsymbol{\hat{f}_s(\alpha)=F_s[f(x)]=\int_0^\infty f(x)\sin\alpha x dx}$

Using $F_s(e^{-ax})$,

$F_s(e^{-ax})=\frac{\alpha}{a^2+\alpha^2}$,

Differentiate w.r.t. $\boldsymbol{a}$ both sides,

$-\frac{2a\alpha}{a^2+\alpha^2}=\int_0^\infty -xe^{-ax}\sin\alpha x\,dx,$

$\Longrightarrow \int_0^\infty xe^{-ax}\sin\alpha x\,dx=\frac{2a\alpha}{a^2+\alpha^2}.$

Exercise:

Find the Fourier Transforms of the following functions :

(1) $f(x)=\begin{cases} x^2, & |x|\le a \\ 0, & |x|>a \end{cases}$ where 'a' is a positive constant.

Answer: $\left(\frac{4a^2}{\lambda}-\frac{4}{\lambda^3}\right)\sin\lambda a+\frac{4ag}{\lambda^2}\cos\lambda a$

(2) $f(x)=\begin{cases} 0, & x<a \\ 1, & a\le x\le b \\ 0, & x>b \end{cases}$ where 'a' and 'b' are positive constants

(3) $f(x)=\begin{cases} 1-|x|, & |x|\le 1 \\ 0, & |x|>1 \end{cases}$

Answer: $\frac{2}{\lambda^2}(1-\cos\lambda)$

(4) Show that Fourier transform of $f(x)=\begin{cases} a^2-x^2, & |x|\le a \\ 0, & |x|>a \end{cases}$ is $\frac{2}{\lambda^3}\sqrt{\frac{2}{\pi}}(\sin\lambda a-\lambda a\cos\lambda a)$.

Hence deduce the following:

i. $\int_0^\infty \frac{\sin x-x\cos x}{x^3}dx=\frac{\pi}{4}$, ii. $\int_0^\infty \frac{\sin x-x\cos x}{x^3}\cos\left(\frac{x}{2}\right)dx=\frac{3\pi}{16}$.

(5) Find the Fourier cosine transform of $f(x)=\begin{cases} x, \text{for } 0<x<1 \\ 2-x, \text{ for } 1<x<2. \\ 0, \text{for } x>2 \end{cases}$

Answer: $\frac{2}{\lambda^2}\cos\lambda-\frac{1}{\lambda^2}\cos 2\lambda-\frac{1}{\lambda^2}$.

(6) Find the Fourier cosine transform of $\frac{e^{-ax}}{x}$.

Answer: $\frac{1}{2}\log(a^2+\lambda^2)+c$.

(7) Find Fourier cosine transform of $\frac{1}{1+x^2}$. Hence derive Fourier sine transform of $\frac{x}{1+x^2}$.

Answer: $F_c[f(x)]=\frac{\pi}{2}e^{-s}$, $F_s[g(x)]=\frac{\pi}{2}e^{-s}$.

CONVOLUTION:

The convolution of two functions f(x) and g(x) over the interval $(-\infty,\infty)$ is defined as

$$f*g=\int_{-\infty}^{\infty} f(u)g(x-u)du=h(x).$$

CONVOLUTION THEOREM FOR FOURIER TRANSFORMS:

The Fourier transform of the convolution of $f(x)$ and $g(x)$ is the product of their Fourier Transforms

$$F\{f(x)*g(x)\}=F\{f(x)\}.F\{g(x)\}.$$

Proof:

$$F\{f(x)*g(x)\}=F\left\{\int_{-\infty}^{\infty} f(u)g(x-u)du\right\},$$

$$=F\left\{\int_{-\infty}^{\infty} f(u)g(x-u)du\right\},$$

$$=\int_{-\infty}^{\infty}\left\{\int_{-\infty}^{\infty} f(u)g(x-u)du\right\}e^{i\alpha x}dx,$$

$$=\int_{-\infty}^{\infty} f(u)\left\{\int_{-\infty}^{\infty} g(x-u)du\right\}e^{i\alpha x}dxdu,$$

$$=\int_{-\infty}^{\infty} f(u)\left\{\int_{-\infty}^{\infty} e^{i\alpha(x-u)}g(x-u)d(x-u)\right\}e^{i\alpha x}du,$$

$$=\int_{-\infty}^{\infty} e^{i\alpha u}f(u)\left\{\int_{-\infty}^{\infty} e^{i\alpha t}g(t)d(t)\right\}du where x-u=t,$$

$$=\int_{-\infty}^{\infty} e^{i\alpha u}f(u)du.F(g(t))=\left\{\int_{-\infty}^{\infty} e^{i\alpha x}f(x)dx\right\}F[g(x)]=F\{f(x)\}.F\{g(x)\}.$$

Example:

1. If $F\{f(t)\} = \frac{1}{(1+i\omega)^2}$ and $f(t) = 0$ for $t < 0$, find $f(t)$ using convolution theorem.

   Solution:

   Given $F\{f(t)\} = \frac{1}{(1+i\omega)^2} = \frac{1}{(1+i\omega)} \cdot \frac{1}{(1+i\omega)}$.

   Let $F(\omega) = G(\omega) = \frac{1}{(1+i\omega)}$.

   Then by convolution theorem

   $F\{f * g\} = F(\omega)G(\omega)$,

   $F^{-1}\{F(\omega)G(\omega)\} = f * g$,

   $f = g = F^{-1}\left\{\frac{1}{(1+i\omega)}\right\} = e^{-t}$,

   $F^{-1}\{F(\omega)G(\omega)\} = f * g = \int_{-\infty}^{\infty} f(u)g(t-u)du = \int_0^t e^{-u}e^{-(t-u)}du = te^{-t}$.

Exercise:

1. If $F\{f(t)\} = \frac{1}{(1+\omega^2)^2}$ and $f(t) = 0$ for $t < 0$, find $f(t)$ using convolution theorem.

PARSEVAL`S IDENTITY FOR FOURIER TRANSFORMS:

If the Fourier Transform of $f(x)$ and $g(x)$ are $F(\alpha)$ and $G(\alpha)$ respectively,

(i) $\frac{1}{2\pi}\int_{-\infty}^{\infty} F(\alpha)\overline{G}(\alpha)d\alpha = \int_{-\infty}^{\infty} f(x)\overline{g}(x)dx,$

(ii) $\frac{1}{2\pi}\int_{-\infty}^{\infty} F(\alpha)\overline{G}(\alpha)d\alpha = \int_{-\infty}^{\infty} \left| f(x) \right|^2 dx.$

Note:

PARSEVAL`S IDENTITY FOR FOURIER COSINE TRANSFORMS AND SINE TRANSFORMS:

(i) $\frac{2}{\pi}\int_0^{\infty} F_c(\alpha)G_c(\alpha)d\alpha = \int_0^{\infty} f(x)g(x)dx$

(ii) $\frac{2}{\pi}\int_0^{\infty} F_s(\alpha)G_s(\alpha)d\alpha = \int_0^{\infty} f(x)g(x)dx$

(iii) $\frac{2}{\pi}\int_0^{\infty} [F_c(\alpha)]^2 d\alpha = \int_0^{\infty} [f(x)]^2 dx$

(iv) $\frac{2}{\pi}\int_0^{\infty} [F_s(\alpha)]^2 d\alpha = \int_0^{\infty} [f(x)]^2 dx$

Examples:

1.Using Parseval`s identity for Fourier Transform prove that

$$\int_0^\infty \frac{t^2 dt}{(t^2+1)^2} = \frac{\pi}{4}$$

Solution:

$F_s(e^{-t}) = \frac{-\alpha}{1+\alpha^2}$,

By Parseval's identity,

$$\frac{2}{\pi}\int_0^\infty [F_s(\alpha)]^2 d\alpha = \int_0^\infty [f(t)]^2 dt,$$

$$\frac{2}{\pi}\int_0^\infty \frac{\alpha^2}{(1+\alpha^2)^2} d\alpha = \int_0^\infty e^{-2t} dt,$$

$$\int_0^\infty \frac{\alpha^2}{(1+\alpha^2)^2} d\alpha = \frac{\pi}{2}\int_0^\infty e^{-2t} dt = \frac{\pi}{4}.$$

Changing α to t

$$\int_0^\infty \frac{t^2 dt}{(t^2+1)^2} = \frac{\pi}{4}.$$

2. Using Parseval's identity for Fourier Transform prove that

$$\int_0^\infty \frac{dt}{(t^2+1)^2} = \frac{\pi}{4}.$$

Solution:

$F_c(e^{-t}) = \frac{1}{1+\alpha^2}$,

By Parseval's identity,

$$\frac{2}{\pi}\int_0^\infty [F_c(\alpha)]^2 d\alpha = \int_0^\infty [f(t)]^2 dt,$$

$$\frac{2}{\pi}\int_0^\infty \frac{d\alpha}{(1+\alpha^2)^2} = \int_0^\infty e^{-2t} dt = \frac{1}{2},$$

$$\int_0^\infty \frac{d\alpha}{(1+\alpha^2)^2} = \frac{\pi}{4},$$

Exercise: Using Parseval's identity for Fourier Transform prove that

(i) $\int_0^\infty \frac{dt}{(a^2+t^2)(b^2+t^2)} = \frac{\pi}{2ab(a+b)}.$

(ii) $\int_0^\infty \frac{\sin at}{t(a^2+t^2)} dt = \frac{\pi}{2}\frac{1-e^{-a^2}}{a^2}.$

APPLICATION OF FOURIER TRANSFORMS TO BOUNDARY VALUE PROBLEMS:

In one dimensional boundary value problems, the partial differential equation can easily transformed into an ordinary differential equation by applying a suitable transform. The required solution is obtained by solving this equation and inverting by means of the complex inversion formula or by any other method.

If in a problem $u(x,t)_{x=0}$ is given then employing infinite sine transform to remove $\frac{\partial^2 u}{\partial x^2}$ from the differential equation.

If in a problem $\left[\frac{\partial u(x,t)}{\partial x}\right]_{x=0}$ is given then employing infinite cosine transform to remove $\frac{\partial^2 u}{\partial x^2}$ from the differential equation.

Example:

Determine the distribution of temperature in the semi-infinite medium $x \geq 0$, when the end $x = 0$ is maintained at zero temperature and the ini8tial distribution of temperature is $f(x)$.

Solution: Let u(x,t) be the temperature at any point x and at any time t.
To solve the heat-flow equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} (x > 0, t > 0) \qquad \text{-----} \qquad \text{(i)}$$

Subject to the initial condition u(x,0) = f(x) ---------(ii)
and the boundary condition u(0,t) = 0----------(iii)

Taking Fourier Sine Transform of (i)

$$\frac{\partial \bar{u}_s}{\partial t} = c^2[\alpha\, u(0,t) - \alpha^2 \bar{u}_s] \;\; or \;\; \frac{\partial \bar{u}_s}{\partial t} + c^2\alpha^2 \bar{u}_s = 0 \text{-----(iv)}$$

Also the Fourier sine transform of (ii) is $\bar{u}_s = \bar{f}_s(\alpha) at\, t = 0.$ -------(v)

Solving (iv) and (v),

$$\bar{u}_s = \bar{f}_s(\alpha) e^{-c^2\alpha^2 t}$$

By inverse Fourier transform

$$u(x,t) = \frac{2}{\pi}\int_0^\infty \bar{f}_s(\alpha) e^{-c^2\alpha^2 t} \sin x\alpha d\alpha.$$

Video Links:

https://nptel.ac.in/courses/111105035/30
https://nptel.ac.in/courses/117101055/16
https://nptel.ac.in/courses/122107037/27
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